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Revisiting Robustness in Graph Machine Learning

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Graph Neural Networks (GNNs) suffer from **over-robustness**

- Common threat models allow perturbations which change the node-level semantic content in graphs
- GNNs are **wrongly robust** to these changes
- Introduce semantic-aware notions of robustness
- No robustness-accuracy tradeoff for inductively classifying a new node

On Adversarial Examples

- Small perturbations (ℓ_p-norm)
- Change prediction of classifier f
- Unchanged semantic content (e.g., image category)

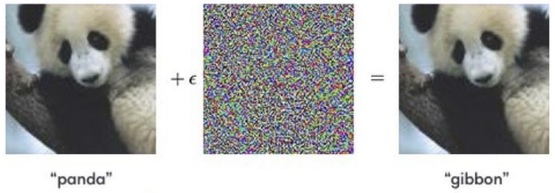
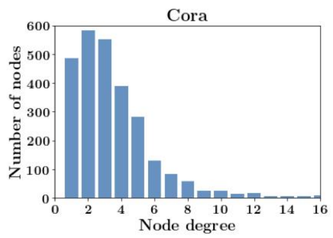


Figure: Goodfellow et al. (2015)

What constitutes a small, semantics-preserving perturbation to a graph?

Commonly: **Restrict** number of inserted / deleted edges (ℓ_0-norm)



⚠ Majority of nodes are low-degree
→ ***Large effect*** on neighborhood

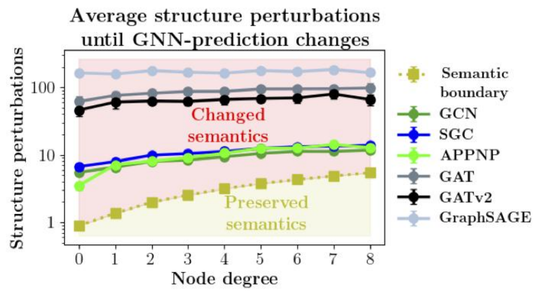
Do common threat models preserve semantic content?

Use Random Graph Models

- Exactly measure (node-level) semantic change

New Phenomenon

Over-robustness: Robustness *beyond* semantic change



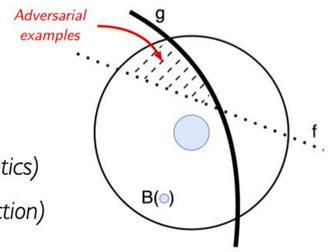
Semantic-Aware Robustness

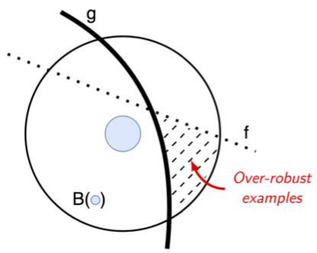
Idea: Use reference node classifier g to indicate semantic change.

Adversarial Example

A perturbed graph $\tilde{G} \in \mathcal{B}(G)$ is said to be *adversarial* for a node classifier f at a target node v w.r.t. g iff:

i. $f(G,y)_v = g(G,y)_v$ *(correct clean prediction)*
ii. $g(\tilde{G},y)_v = g(G,y)_v$ *(perturbation preserves semantics)*
iii. $f(\tilde{G},y)_v \neq g(G,y)_v$. *(node classifier changes prediction)*





Over-robust Example

A perturbed graph $\tilde{G} \in \mathcal{B}(G)$ is said to be an *over-robust example* for a node classifier f at a target node v w.r.t. g iff:

i. $f(G,y)_v = g(G,y)_v$ *(correct clean prediction)*
ii. $g(\tilde{G},y)_v \neq g(G,y)_v$ *(perturbation changes semantics)*
iii. $f(\tilde{G},y)_v = g(G,y)_v$. *(node classifier stays unchanged)*

Theorem (informal)

There is no semantic-aware robustness-accuracy tradeoff for non-i.i.d. graph data when classifying a newly added node to a graph.

Experiments

Setup: Contextual stochastic block models (CSBM)

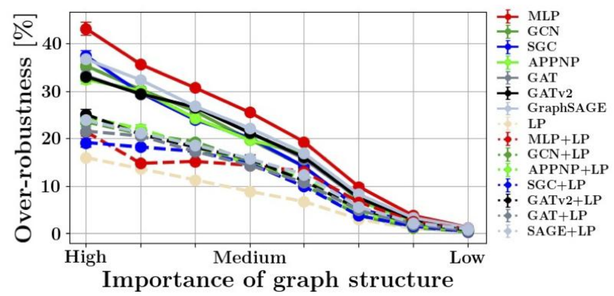
- *Derived* Bayes optimal classifier as reference classifier g

Table: Existence probability of graphs with changed semantics.

Threat models	Importance of graph structure: High	·	·	Medium	·	·	Low	·
$\mathcal{B}_2(\cdot)$	36%	31%	26%	20%	14%	6%	2%	1%
$\mathcal{B}_{\text{deg}}(\cdot)$	**76%**	**60%**	**55%**	**49%**	40%	22%	9%	3%
$\mathcal{B}_{\text{deg+2}}(\cdot)$	100%	100%	99%	93%	81%	52%	25%	9%

→ **High prevalence** of graphs with changed semantics!

How much of the measured robustness is in fact over-robustness?



→ **Significant amount** of conventional robustness is **over-robustness**

Take-Aways

- Use strong attacks
  → Reduce the amount of over-robustness measured
- Be cautious in applying ℓ_0-norm restricted adversaries
- **More research** into semantic-preserving perturbations **necessary**